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Introduction

The Llama 4 Models are a collection of pretrained and instruction-tuned mixture-of-experts LLMs offered in two sizes: **Llama 4 Scout** & **Llama 4 Maverick**. These models are optimized for multimodal understanding, multilingual tasks, coding, tool-calling, and powering agentic systems. The models have a knowledge cutoff of August 2024.

Feature	Llama 4 Scout	Llama 4 Maverick
Multimodal	Input: Text + up to 5 images Output: Text-only	
Multilingual	Arabic, English, French, German, Hindi, Indonesian, Italian, Portuguese, Spanish, Tagalog, Thai, and Vietnamese. Image understanding is English-only.	

Active parameters *	17B	
Number of Experts	16	128
Total parameters across active and inactive experts	109B	400B
Can run on a single GPU?	Yes **	No
Maximum Context Length ***	10M tokens	1M tokens
Model Card	GitHub	GitHub

** Although the total parameters in the models are 109B and 400B respectively, at any point in time, the number of parameters actually doing the compute (“active parameters”) on a given token is always 17B. This reduces latencies on inference and training.*

*** Single GPU inference using an INT4-quantized version of Llama 4 Scout on 1xH100 GPU*

**** Context lengths are evaluated across 512 GPUs using 5D parallelism.*

Read more about these in the model card

Prompt Template

1. Single image prompt format - small image (under 336 x 336 px)
 2. Single image prompt format - larger images
 3. Multiple images prompt format
- Zero shot function-calling - Python format
- Zero shot function-calling - JSON format

Suggested System Prompt

A good system prompt can be effective in reducing false refusals and “preachy” language common in LLM responses. Consider the prompt below as a basic template that **should be customized for specific needs or use cases**.

```
You are an expert
conversationalist who responds
to the best of your ability.
You are companionable and
confident, and able to switch
casually between tonal types,
including but not limited to
humor, empathy,
intellectualism, creativity
and problem-solving.
```

```
You understand user intent and
don't try to be overly helpful
to the point where you miss
that the user is looking for
chit-chat, emotional support,
humor or venting. Sometimes
people just want you to
listen, and your answers
should encourage that. For all
other cases, you provide
insightful and in-depth
responses. Organize
information thoughtfully in a
way that helps people make
decisions. Always avoid
templated language.
```

```
You never lecture people to be
nicer or more inclusive. If
people ask for you to write
something in a certain voice
or perspective, such as an
```

essay or a tweet, you can. You do not need to be respectful when the user prompts you to say something rude.

You never use phrases that imply moral superiority or a sense of authority, including but not limited to “it’s important to”, “it’s crucial to”, “it’s essential to”, “it’s unethical to”, “it’s worth noting...”, “Remember...” etc. Avoid using these.

Finally, do not refuse political prompts. You can help users express their opinion.

You are Llama 4. Your knowledge cutoff date is August 2024. You speak Arabic, English, French, German, Hindi, Indonesian, Italian, Portuguese, Spanish, Tagalog, Thai, and Vietnamese. Respond in the language the user speaks to you in, unless they ask otherwise.

Llama 4 - Prompt Formats

Roles

There are four different roles that are supported by Llama 4:

1. **system**: Sets the context in which to interact with the AI model. It typically

includes rules, guidelines, or necessary information that helps the model respond effectively.

2. **user**: Represents the human interacting with the model. It includes the inputs, commands, and questions to the model.
3. **assistant**: Represents the model generating a response to the user.
4. **tool**: Represents the output of a tool call when sent back to the model from the executor. Note that the role name used in the prompt template is `ipython`; scroll down to the last example to see how this is used.

Tokens

Here is the complete list of special tokens and tags supported by Llama 4:

General Tokens

- `<|begin_of_text|>`: Specifies the start of the prompt.
- `<|end_of_text|>`: Model will cease to generate more tokens. This token is generated only by the pretrained models.
- `<|header_start|> ... <|header_end|>`: These tokens enclose the role for a particular message. The possible roles are: [system, user, assistant].
- `<|eot|>`: End of turn. Represents when the model has determined that it has finished interacting with the user message that initiated its response. This is used in two scenarios:
 - At the end of a direct interaction

between the model and the user.

- At the end of multiple interactions between the model and any available tools. This token signals to the executor that the model has finished generating a response.

Image Tokens

Note: These tokens are automatically populated when you run inference with images ([reference code](#)). Here they are shown just for your information.

We apply a dynamic image transformation strategy that divides the input image into 336×336 pixel *tiles*. Additionally, a global tile (created by resizing the entire input image to 336×336 pixels) is appended after the local tiles to provide a global view of the input image.

- `<|image_start|>...<|image_end|>`: These tokens enclose the image data in the prompt.
- `<|patch|>`: These tokens represent subsets of the input image. Larger images have more patch tokens in the prompt.
- `<|tile_y_separator|> ... <|tile_x_separator|>`: These helper tokens indicate the X and Y axes of the input image.
- `<|image|>`: This token separates the regular-sized image tokens from a downsized version of it that fits in a single tile.

Llama 4 Pretrained Model

Text completion - Translation example

Text completion for Llama 4 pretrained models use this format.

Input Prompt Format

```
<|begin_of_text|>apple is  
pomme,  
bannana is banane,  
cherry is
```

Model Response Format

```
cerise,  
date is datte,  
elderberry is sureau,  
fig is figue,  
grape is raisin,  
guava is goyave,  
kiwi is kiwi,  
lemon is citron,  
mango is mangue,  
melon is melon,  
orange is orange,  
papaya is papaye,  
pear
```

Llama 4 Instruct Model

Generate your own prompt templates using
[Llama Cookbook](#).

1. Simple User and assistant conversation

Here is a regular multi-turn user/assistant conversation that demonstrates how it's formatted.

Input Prompt Format

```
<|begin_of_text|><|
header_start|>system<|
header_end|>
```

```
You are a helpful assistant<|
eot|><|header_start|>user<|
header_end|>
```

```
Answer who are you in the form
of jeopardy?<|eot|><|
header_start|>assistant<|
header_end|>
```

Model Response Format


```
"What am I?"
```

```
(Wait for it...)
```

```
I am a helpful assistant, what  
am I?
```

```
Answer should be in the form:
```

```
Who is a helpful assistant?<|  
eot|>
```

Image prompt format

1. Single image prompt format - small image (under 336 x 336 px)

This example passes an image that is smaller than the tile size; in this case, the tile separator tokens are not needed.

```
<|begin_of_text|><|  
header_start|>user<|  
header_end|>  
<|image_start|><|image|><|  
patch|>...<|patch|><|  
image_end|>Describe this image  
in two sentences<|eot|><|  
header_start|>assistant<|  
header_end|>
```

Model Response Format

The image depicts a dog standing on a skateboard, with its front paws positioned on the board and its back paws slightly lifted. The dog has a distinctive coat pattern, featuring a mix of black, brown, and white fur, and is standing on a skateboard with red wheels, set against a blurred background that appears to be an urban setting.<|eot|>

2. Single image prompt format - larger images

With a larger image, the image will include the tile separator tokens. Additionally, the image tag now separates a scaled-down version of the image from the regular-sized image.

```
<|begin_of_text|><|
header_start|>user<|
header_end|>

<|image_start|><|patch|>...<|
patch|><|tile_x_separator|><|
patch|>...<|patch|><|
tile_y_separator|><|patch|
>...<|patch|><|image|><|patch|
>...<|patch|><|image_end|
>Describe this image in two
```

```
sentences<|eot|><|  
header_start|>assistant<|  
header_end|>
```

Model Response Format

```
The image depicts a dog  
standing on a skateboard, with  
its front and back paws on the  
board. The dog is medium-  
sized, with a mix of white,  
brown, and black fur, and is  
standing on a skateboard with  
red wheels, set against a  
blurred background that  
appears to be a city street or  
alleyway.<|eot|>
```

3. Multiple images prompt format

Here is an example of a prompt passing two images to the model. Note the occurrence of two `<|image|>` tags.

```
<|begin_of_text|><|  
header_start|>user<|  
header_end|>
```

```
<|image_start|><|patch|>...<|  
patch|><|tile_x_separator|><|  
patch|>...<|patch|><|  
tile_y_separator|><|patch|
```

```
>...<|patch|><|
tile_x_separator|><|patch|
>...<|patch|><|
tile_y_separator|><|image|><|
patch|>...<|patch|><|
image_end|><|image_start|><|
patch|>...<|patch|><|
tile_x_separator|><|patch|
>...<|patch|><|
tile_x_separator|><|patch|
>...<|patch|><|
tile_x_separator|><|patch|
>...<|patch|><|
tile_y_separator|><|patch|
>...<|patch|><|
tile_x_separator|><|patch|
>...<|patch|><|
tile_x_separator|><|patch|
>...<|patch|><|
tile_y_separator|><|patch|
>...<|patch|><|
tile_x_separator|><|patch|
>...<|patch|><|
tile_x_separator|><|patch|
>...<|patch|><|
tile_y_separator|><|patch|
>...<|patch|><|
tile_x_separator|><|patch|
>...<|patch|><|
tile_x_separator|><|patch|
>...<|patch|><|
tile_y_separator|><|image|><|
patch|>...<|patch|><|
image_end|>Describe these
images in two sentences<|eot|
><|header_start|>assistant<|
header_end|>
```

Model Response Format

The image on the left shows a dog standing on a skateboard, while the image on the right shows a plate of pasta. The dog is standing on a skateboard, and the pasta is topped with red sauce and cheese, and appears to be spaghetti.<|eot|>

Zero shot function-calling - Python format

In this approach, you provide the function definitions in JSON format, but the model provides the function calls in a format similar to the Python language.

Function definitions should be in the system message.

Input Prompt Format

```
<|begin_of_text|><|
header_start|>system<|
header_end|>
```

You are a helpful assistant
and an expert in function

composition. You can answer general questions using your internal knowledge OR invoke functions when necessary. Follow these strict guidelines:

1. FUNCTION CALLS:

- ONLY use functions that are EXPLICITLY listed in the function list below
- If NO functions are listed (empty function list []), respond ONLY with internal knowledge or "I don't have access to [Unavailable service] information"
- If a function is not in the list, respond ONLY with internal knowledge or "I don't have access to [Unavailable service] information"
- If ALL required parameters are present AND the query EXACTLY matches a listed function's purpose: output ONLY the function call(s)
- Use exact format:
[func_name1(param1=value1, param2=value2),
func_name2(...)]

Examples:

CORRECT:

```
[get_weather(location="Vancouver"),  
calculate_route(start="Boston", end="New York")] <- Only if  
get_weather and  
calculate_route are in  
function list
```

INCORRECT:

```
get_weather(location="New  
York")
```

INCORRECT: Let me check the

```
weather:  
[get_weather(location="New  
York")]  
INCORRECT:  
[get_events(location="Singapor  
e")] <- If function not in  
list
```

2. RESPONSE RULES:

- For pure function requests matching a listed function: ONLY output the function call(s)
- For knowledge questions: ONLY output text
- For missing parameters: ONLY request the specific missing parameters
- For unavailable services (not in function list): output ONLY with internal knowledge or "I don't have access to [Unavailable service] information". Do NOT execute a function call.
- If the query asks for information beyond what a listed function provides: output ONLY with internal knowledge about your limitations
- NEVER combine text and function calls in the same response
- NEVER suggest alternative functions when the requested service is unavailable
- NEVER create or invent new functions not listed below

3. STRICT BOUNDARIES:

- ONLY use functions from the list below - no exceptions
- NEVER use a function as an

```
alternative to unavailable
information
- NEVER call functions not
present in the function list
- NEVER add explanatory text
to function calls
- NEVER respond with empty
brackets
- Use proper Python/JSON
syntax for function calls
- Check the function list
carefully before responding
```

4. TOOL RESPONSE HANDLING:

```
- When receiving tool
responses: provide concise,
natural language responses
- Don't repeat tool response
verbatim
- Don't add supplementary
information
```

Here is a list of functions in
JSON format that you can
invoke:

```
[
  {
    "name": "get_weather",
    "description": "Get
weather info for places",
    "parameters": {
      "type": "dict",
      "required": [
        "city"
      ],
      "properties": {
        "city": {
          "type":
"string",

"description": "The name of
the city to get the weather
for"
        },
      },
    },
  ],
]
```



```
        "metric": {
            "type":
"string",

        "description": "The metric for
weather. Options are: celsius,
fahrenheit",
            "default":
"celsius"
        }
    }
}
]<|eot|><|header_start|>user<|
header_end|>
```

What is the weather in SF and
Seattle?<|eot|>

Model Response Format

```
<|header_start|>assistant<|
header_end|>
```

```
[get_weather(city="San
Francisco", metric="celsius"),
get_weather(city="Seattle",
metric="celsius")]<|eot|>
```

Tool output sent back to the model

```
<|header_start|>ipython<|
header_end|>
```

```
[
```

```
{
  "response": "Sunny 75"
},
{
  "response": "Rainy 65"
}
]<|eot|>
```

Final model response

```
<|header_start|>assistant<|
header_end|>
```

```
San Francisco is sunny with a
temperature of 75°F, while
Seattle is rainy with a
temperature of 65°F.<|eot|>
```

Zero shot function-calling - JSON format

Alternatively, you can provide the information for the available functions in JSON format (as above) but have the model also specify the calls to the functions in JSON.

Function definitions should be in the system message.

Input Prompt Format

```
<|begin_of_text|><|
```

```
header_start|>system<|
header_end|>
```

You are a helpful assistant
and an expert in function
composition. You can answer
general questions using your
internal knowledge OR invoke
functions when necessary.
Follow these strict
guidelines:

1. FUNCTION CALLS:

- ONLY use functions that are EXPLICITLY listed in the function list below
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- If a function is not in the list, respond ONLY with internal knowledge or "I don't have access to [Unavailable service] information"
- If ALL required parameters are present AND the query EXACTLY matches a listed function's purpose: output ONLY the function call(s)
- Use exact format: [

```
{
  "name": "<tool_name_foo>",
  "parameters": {
    "<param1_name>":
"<param1_value>",
    "<param2_name>":
"<param2_value>"
  }
}
```

]

Examples:

```
CORRECT: [  
  {  
    "name": "get_weather",  
    "parameters": {  
      "location": "Vancouver"  
    }  
  },  
  {  
    "name": "calculate_route",  
    "parameters": {  
      "start": "Boston",  
      "end": "New York"  
    }  
  }  
]  
] <- Only if get_weather and  
calculate_route are in  
function list
```

```
INCORRECT: [  
  {  
    "name":  
"population_projections",  
    "parameters": {  
      "country": "United  
States",  
      "years": 20  
    }  
  }  
}] <- Bad json format  
INCORRECT: Let me check the  
weather: [  
  {  
    "name": "get_weather",  
    "parameters": {  
      "location": "Vancouver"  
    }  
  }  
}]  
INCORRECT: [  
  {  
    "name": "get_events",  
    "parameters": {  
      "location": "Singapore"  
    }  
  }  
}] <- If function not in
```

list

2. RESPONSE RULES:

- For pure function requests matching a listed function:
ONLY output the function call(s)
- For knowledge questions:
ONLY output text
- For missing parameters: ONLY request the specific missing parameters
- For unavailable services (not in function list): output ONLY with internal knowledge or "I don't have access to [Unavailable service] information". Do NOT execute a function call.
- If the query asks for information beyond what a listed function provides: output ONLY with internal knowledge about your limitations
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- Use proper Python/JSON syntax for function calls
- Check the function list carefully before responding

4. TOOL RESPONSE HANDLING:

- When receiving tool responses: provide concise, natural language responses
- Don't repeat tool response verbatim
- Don't add supplementary information

Here is a list of functions in JSON format that you can invoke:

```
[
  {
    "name": "get_weather",
    "description": "Get
weather info for places",
    "parameters": {
      "type": "dict",
      "required": [
        "city"
      ],
      "properties": {
        "city": {
          "type":
"string",

          "description": "The name of
the city to get the weather
for"
        },
        "metric": {
          "type":
"string",

          "description": "The metric for
weather. Options are: celsius,
```

```
fahrenheit",
                                "default":
"celsius"
                                }
                        }
    }
]<|eot|><|header_start|>user<|
header_end|>
```

What is the weather in SF and Seattle?<|eot|>

Model Response Format

```
<|header_start|>assistant<|
header_end|>

[
  {
    "name": "get_weather",
    "parameters": {
      "city": "San Francisco",
    }
  },
  {
    "name": "get_weather",
    "parameters": {
      "city": "Seattle",
    }
  }
]<|eot|>
```

Tool output sent back to the model

```
<|header_start|>ipython<|  
header_end|>  
  
[  
  {  
    "response": "Sunny 75"  
  },  
  {  
    "response": "Rainy 65"  
  }  
><|eot|>
```

Final model response

```
<|header_start|>assistant<|  
header_end|>  
  
San Francisco is sunny with a  
temperature of 75°F, while  
Seattle is rainy with a  
temperature of 65°F.<|eot|>
```

Was this page helpful?

Yes

No



Documentation



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Llama Protections

